determination, he adds that this is ‘I think making a difference in the areas where I want to be’.

When Passion and Philanthropy Converge

Ratan has also turned his attention to a matter close to his heart, the future of cars as we know it. He has been investigating if the internal combustion engine (ICE), which has improved over decades of research and development, can survive if it does not emit noxious fumes. He is grappling with the question if alternate fuel can drive the vehicles and has been making attempts to see if hydrogen can be used as a fuel. If this works, the nearly 2 billion ICE cars that run today might not need to be given up only because of pollution.

Thinking ahead of his time, Ratan says that making 100 per cent electric cars can have several problems. There is the issue of charging cars in the remotest parts of the country, among others. On the other hand, shifting to hydrogen would not be easy. It is not that he has not thought on those lines.

He says, ‘I was at MIT when a particular scientist made a presentation of splitting of water vapour into hydrogen and oxygen. He was talking of photosynthesis, synthetic photosynthesis and I thought this is the same thing as what we are looking at so I went to him, I said if we use your catalyst and split water into hydrogen and oxygen, could you feed that into an engine? And he said yes, and I, in turn, sanctioned the investment of 8 million US dollars into the programme. Unfortunately, I gave it to Tata Industries which started looking at a particular kind of battery that would run on hydrogen and moved it towards hydrogen cells for cars, meant for electric versions.’

This killed the project as the scientist was looking for electric power in cars to do the splitting. ‘We got it working for a little model aircraft engine and could use the catalyst that could produce oxygen which you see in the lab going up in bubbles and on the other side, produce hydrogen,’ Ratan says. This was ‘proof of concept that we could do it’, he adds ruefully. The question was to get it in quantities as required. This led to him looking at what was called a new generation fuel containing hydrogen and ammonia. But it could have led to a much ‘bigger thing if it had happened’, he remarks.